R120 | Same-checkpoint VAL / TEST

split	group	task_count	task_macro_any_best_top1_pct	task_macro_mixed_top1_pct	task_macro_train_scaled_excess
VAL	ALL24	24	73.19185	73.08259	0.00507
VAL	NONROUTING6	6	79.43037	79.43037	0.00833
VAL	ROUTING18	18	71.11235	70.96667	0.00398
TEST	ALL24	24	73.30790	73.19957	0.00508
TEST	NONROUTING6	6	79.10198	79.10198	0.00865
TEST	ROUTING18	18	71.37654	71.23210	0.00388

ALL24 is task-macro; raw costs/profits/makespans are never pooled. Routing native Top1 is separate. New-task ties use any-best Top1. TEST/LIB did not select this checkpoint. This is a new multidomain baseline; historical R94B is not a matched architectural control.

R120 | VAL native objectives

problem	objective_direction	top1_pct	mean_objective	sbs_objective	oracle_objective	actual_regret_pct	train_scaled_excess
ATSP	min	75.91111	1.63896	1.67690	1.63247	0.39571	0.00386
BPP	min	98.10000	104.47600	104.47600	104.45700	0.03433	0.00018
CVRP	min	46.71111	18.16489	18.26938	18.06409	0.74630	0.00555
FJSP	min	43.00000	368.25300	375.32400	355.99000	3.01645	0.03261
JSSP	min	58.80000	1665.87600	1665.87600	1643.77500	1.68703	0.01329
MIS	max	87.90000	75.01300	74.16500	75.20000	0.33421	0.00252
MKP	max	93.55556	119673.56667	119673.56667	119701.34667	0.09069	0.00023
OVRP	min	40.91111	7.46147	7.46958	7.41578	0.62926	0.00613
OVRPB	min	88.44444	6.35816	6.35725	6.34108	0.27838	0.00268
OVRPBL	min	89.33333	6.36932	6.37009	6.35282	0.27044	0.00259
OVRPBLTW	min	83.22222	10.61024	10.60996	10.57130	0.34649	0.00366
OVRPBTW	min	83.57778	10.66052	10.66110	10.62324	0.32419	0.00351
OVRPL	min	40.91111	7.43931	7.44571	7.39042	0.68310	0.00656
OVRPLTW	min	83.44444	10.15771	10.15742	10.12982	0.28111	0.00276
OVRPTW	min	83.88889	10.14280	10.14373	10.11686	0.25339	0.00256
SMTWTP	min	95.22667	105683.30347	105745.63840	105565.35787	8.48382	0.00112
TSP	min	46.28889	8.02532	8.04535	7.99053	0.42542	0.00433
VRPB	min	80.91111	9.35542	9.35488	9.31875	0.40404	0.00392
VRPBL	min	78.77778	9.37057	9.37152	9.32837	0.45770	0.00449
VRPBLTW	min	79.51111	19.79487	19.79704	19.70642	0.43703	0.00447
VRPBTW	min	78.64444	19.94727	19.95043	19.85108	0.46478	0.00487
VRPL	min	37.95556	12.43184	12.43796	12.38344	0.39829	0.00390
VRPLTW	min	78.82222	18.76459	18.76587	18.70784	0.31850	0.00302
VRPTW	min	80.13333	18.81886	18.81943	18.76614	0.29652	0.00280

SBS fitted on TRAIN. Relative regret is defined only for strictly positive native pool-oracles; zero-oracle count and absolute losses are in metrics.csv. Canonical tie display is not a unique ground truth.

R120 | TEST native objectives

problem	objective_direction	top1_pct	mean_objective	sbs_objective	oracle_objective	actual_regret_pct	train_scaled_excess
ATSP	min	75.64444	1.63577	1.67421	1.62944	0.38913	0.00377
BPP	min	99.10000	104.36100	104.36100	104.35200	0.01684	0.00009
CVRP	min	47.82222	18.01681	18.12498	17.91851	0.72570	0.00542
FJSP	min	42.80000	368.51100	375.68800	355.57600	3.09254	0.03440
JSSP	min	59.90000	1661.13100	1661.13100	1639.38500	1.66598	0.01308
MIS	max	85.60000	75.05700	74.21000	75.27200	0.39750	0.00290
MKP	max	91.85185	119645.91407	119645.91407	119676.13481	0.09271	0.00025
OVRP	min	40.62222	7.44703	7.45575	7.40049	0.63981	0.00624
OVRPB	min	88.17778	6.37061	6.37083	6.35223	0.29603	0.00289
OVRPBL	min	89.28889	6.37560	6.37574	6.35931	0.26721	0.00256
OVRPBLTW	min	83.60000	10.66193	10.66215	10.62557	0.31895	0.00342
OVRPBTW	min	84.91111	10.65885	10.65959	10.62475	0.29301	0.00321
OVRPL	min	41.82222	7.46416	7.47254	7.41803	0.63382	0.00619
OVRPLTW	min	82.93333	10.08569	10.08611	10.05900	0.26619	0.00264
OVRPTW	min	83.93333	10.09413	10.09306	10.06814	0.25682	0.00257
SMTWTP	min	95.36000	105602.94373	105679.28133	105476.72453	15.50918	0.00119
TSP	min	46.64444	7.97095	7.98962	7.93759	0.41178	0.00416
VRPB	min	80.68889	9.34970	9.35181	9.31530	0.37285	0.00368
VRPBL	min	79.64444	9.39540	9.39638	9.35597	0.42673	0.00419
VRPBLTW	min	79.68889	19.70629	19.70707	19.61328	0.45440	0.00470
VRPBTW	min	78.77778	19.66638	19.66573	19.57767	0.45048	0.00450
VRPL	min	37.57778	12.35028	12.35614	12.30096	0.41038	0.00398
VRPLTW	min	79.28889	18.80231	18.80314	18.74514	0.32043	0.00305
VRPTW	min	81.11111	18.68927	18.69055	18.63741	0.28964	0.00276

SBS fitted on TRAIN. Relative regret is defined only for strictly positive native pool-oracles; zero-oracle count and absolute losses are in metrics.csv. Canonical tie display is not a unique ground truth.

R120 | Historical NSS reference

dataset	nss_top1_pct	current_top1_pct	nss_mean_cost	current_mean_cost	status
TSP TEST	45.88889	46.64444	7.97040	7.97095	same EVAL costs/winners/pool; hist
CVRP TEST	45.97778	47.82222	18.00480	18.01681	same EVAL costs/winners/pool; hist
TSPLIB	20.40816	24.48980	8.21791	8.21240	same EVAL costs/winners/pool; hist
CVRPLIB	70.00000	57.00000	67.10111	67.40262	same EVAL costs/winners/pool; hist

Only same-population evaluation labels and candidates are compared. Missing references remain missing. This does not match the training populations: R120 routing uses 81,000 TRAIN instances/task.
NSS weights: R79 historical VALID-selected NSS weights; not retrained on R84/new90k
Historical NSS TRAIN: /public/home/shiys/selection_data_new_only_B_filtered_single_reld_mt_seed2_20261014; rows/task=70114

R120 | Zero-choice methods with unique wins or positive replacement loss (1-25)

problem	solver	unique wins	unique missed	unique recall	unique missed loss / s_t	tied replacement excess
ATSP	GLOP	10	10	0.00000	0.22670	0.00000
ATSP	MATPOENET	307	307	0.00000	5.01056	0.08069
BPP	BPP/BFD/deterministic_v1	0	0	missing	0.00000	9.00000
BPP	BPP/FFD/deterministic_v1	0	0	missing	0.00000	9.00000
CVRP	RouteFinder	67	67	0.00000	0.51974	0.00000
FJSP	FJSP/DAN/10x5/greedy_repe	82	82	0.00000	1.65937	62.00000
FJSP	FJSP/DAN/15x10/greedy_rep	60	60	0.00000	1.49716	81.00000
FJSP	FJSP/MWKR/non_delay_id_t	6	6	0.00000	0.09573	0.00000
JSSP	JSSP/L2D/10_10_1_99/greed	42	42	0.00000	1.43811	741.00000
JSSP	JSSP/L2D/15_15_1_99/greed	52	52	0.00000	1.41826	757.00000
JSSP	JSSP/L2D/20_20_1_99/greed	78	78	0.00000	2.12739	264.00000
JSSP	JSSP/L2D/30_20_1_99/greed	50	50	0.00000	1.51389	803.00000
JSSP	JSSP/SPT/non_delay_id_ties	154	154	0.00000	5.85228	58.00000
MIS	MIS/DIFUSCO_ER/native_buc	19	19	0.00000	0.45906	2.00000
MKP	MKP/AntSystem/deepaco_na	12	12	0.00000	0.03462	0.00000
MKP	MKP/SurrogateGreedy/deter	98	98	0.00000	0.30635	0.00000
OVRP	MTPOMO	44	44	0.00000	0.36635	0.00000
OVRP	MVMOE	24	24	0.00000	0.27810	0.00000
OVRPBL	MTPOMO	219	219	0.00000	5.80835	0.00000
OVRPL	MTPOMO	58	58	0.00000	0.71325	0.00000
OVRPL	MVMOE	34	34	0.00000	0.38213	0.00000
OVRPLTW	MoSES_CaDA	30	30	0.00000	0.63097	0.00000
OVRPLTW	MoSES_RF	5	5	0.00000	0.10727	0.00000
OVRPLTW	RouteFinder	11	11	0.00000	0.47992	0.00000
OVRPTW	MoSES_CaDA	26	26	0.00000	0.39267	0.00000

Existing false_negative/recall/positive_winner_support use all-best support, including ties. Unique-best recall and directed replacement loss measure missed opportunities. Selecting another exact best has zero replacement loss; tied methods receive no selection quotas. Per-method tied losses can overlap.

R120 | Zero-choice methods with unique wins or positive replacement loss (26-39)

problem	solver	unique wins	unique missed	unique recall	unique missed loss / s_t	tied replacement excess
OVRPTW	MoSES_RF	6	6	0.00000	0.12098	0.00000
OVRPTW	RouteFinder	10	10	0.00000	0.28656	0.00000
TSP	DIFUSCO	276	276	0.00000	1.92744	0.27516
TSP	DIFUSCO500	33	33	0.00000	0.15435	0.14159
TSP	LEHD	762	762	0.00000	7.13242	0.05575
TSP	OMNI	28	28	0.00000	0.13873	0.00000
VRPBLTW	MTPOMO	3	3	0.00000	0.08762	0.00000
VRPBTW	MTPOMO	8	8	0.00000	0.16141	0.00000
VRPL	MTPOMO	149	149	0.00000	1.08519	0.00000
VRPL	MVMOE	155	155	0.00000	1.24678	0.00000
VRPLTW	MoSES_RF	9	9	0.00000	0.20275	0.00000
VRPLTW	RouteFinder	19	19	0.00000	0.48161	0.00000
VRPTW	MoSES_RF	5	5	0.00000	0.20568	0.00000
VRPTW	RouteFinder	13	13	0.00000	0.27164	0.00000

Existing false_negative/recall/positive_winner_support use all-best support, including ties. Unique-best recall and directed replacement loss measure missed opportunities. Selecting another exact best has zero replacement loss; tied methods receive no selection quotas. Per-method tied losses can overlap.